Reg. No. | | | | | | | | | | | [12AE81]

EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE – 2016

AIRCRAFT SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Discuss the reliability parameters in maintenance.		07
	b.	Explain the functioning of a fly-by-wire system.		07
2.		Write short notes on:		
		i.	Typical start sequence of a turbojet engine	03
		ii.	Multi-function display unit or glass cock pit	04
		iii.	High energy ignition system	04
		iv.	Wet cycle and dry cycle in engine starts	03
3.		Discuss in brief the following		
		i.	use of fuel as heat sink	03
		ii.	Fuel jettison	03
		iii.	External fuel tanks	04
		iv.	In-flight refueling	04
4.		Explain the following:		
		i.	Advantages of hydraulic system	04
		ii.	The important parameters that affect the design of the hydraulic system.	05
		iii.	Positive displacement rotary swash plate hydraulic pumps.	05
5.		Explain the following:		
		i.	DC power generation	04
		ii.	AC power generation	05
		iii.	Protection functions in a DC system generation control.	05
6.		Write short notes on:		
		i.	Anti-icing/de-icing methods	05
		ii.	Methods of thrust reversing employed in Aircraft	05
		iii.	Vertical speed indicator (VSI)	04
7.		Write short notes on:		
		i.	Humidity control in aircraft	05
		ii.	G tolerance	04
		iii.	Vapour cycle refrigeration systems	05
8.	a.	Explain in detail the Failure Mode Effect Analysis (FMEA) process in system design and development.		07
	b.	Discuss in detail with a neat diagram, a typical aircraft product life cycle in the development process.		07